

MSE 590500 Spring 2026 final exam, written part

Date: 6/8/2026 (Mon), in-class.

Your name: _____

Part I, True or False, circle your choice. 30 points

1. (T) (F) FFT converts a signal from the time domain to the frequency domain.
2. (T) (F) Median filtering is generally effective for removing salt-and-pepper noise (Black pixels (pepper), White pixels (salt)).
3. (T) (F) In OOP, inheritance allows a class to acquire properties from another class.
4. (T) (F) A high R^2 value guarantees that a regression model captures the corresponding physical meanings.
5. (T) (F) KNN is considered a lazy learning algorithm.
6. (T) (F) Perceptron can solve non-linearly separable classification problems.
7. (T) (F) Logistic regression outputs probabilities between 0 and 1.
8. (T) (F) ANNs generally perform better than CNNs in image analysis.
9. (T) (F) Train-test split is used to estimate model generalization performance.
10. (T) (F) A confusion matrix is used for presenting regression performance.
11. (T) (F) PINNs incorporate governing equations into the loss function.
12. (T) (F) Large language models are typically trained using neural networks.
13. (T) (F) In FFT, low-frequency components often correspond to smooth image features.
14. (T) (F) K-means clustering works with labeled data.
15. (T) (F) An ANN may contain multiple hidden layers.

Part II, Single selection, circle your choice. 20 points

1. In signal or image processing, what does FFT stand for?
 - A. Fast Feature Transform
 - B. Fast Fourier Transform
 - C. Fourier Feature Technique
 - D. Feature Frequency Transform
2. What is the primary purpose of train-test splitting?
 - A. Reduce memory usage
 - B. Improve plotting speed
 - C. Estimate model performance on unseen data
 - D. Increase dataset size
3. Which metric is commonly used for regression?
 - A. Accuracy
 - B. Precision

- C. Recall
 - D. R^2
4. In KNN classification, a sample is assigned to:
 - A. The nearest centroid
 - B. The majority class among nearest neighbors
 - C. A random class
 - D. The largest class in the dataset
 5. Which process introduces nonlinearity into neural networks?
 - A. Matrix multiplication
 - B. ReLU activation function
 - C. Dot product
 - D. Gradient descent
 6. The decision boundary in logistic regression is defined by:
 - A. $wx + b = 0$
 - B. $y = x$
 - C. $R^2 = 1$
 - D. Loss = 0
 7. Which algorithm is most appropriate for image classification?
 - A. CNN
 - B. Linear regression
 - C. K-means
 - D. FFT
 8. What does a confusion matrix summarize?
 - A. Fourier coefficients
 - B. Classification results
 - C. Regression residuals
 - D. Neural network weights
 9. PINNs differ from ordinary neural networks because they
 - A. Use no training data
 - B. Include physical equations in the loss function
 - C. Regression residuals
 - D. Neural network weights
 10. A classifier has the following results: true positive = 90, true negative = 80, False positive = 10, and False negative = 20. A quantity is calculated as

$$\frac{90 + 80}{90 + 10 + 80 + 20} = 0.85$$

What does this value mean?

- A. Accuracy
- B. Precision
- C. Recall
- D. R^2

Part III, Multiple selection, circle all your choices. 16 points

1. Which are common image denoising methods?
 - A. Gaussian filter
 - B. Median filter
 - C. Bilateral filters
 - D. Gradient descents
2. Which are supervised learning algorithms?
 - A. K-means
 - B. KNN
 - C. Logistic regression
 - D. SVM
3. Which are common regression metrics?
 - A. R^2
 - B. MAE
 - C. MSE
 - D. Precision
4. Which statements about LLMs are true?
 - A. They are based on neural networks
 - B. They often use transformers
 - C. They can generate text
 - D. They require no training

Part IV, Short answer questions, 36 points

1. Explain how FFT is used in image denoising.

2. Briefly explain a class in OOP.

3. Explain what do the following mean.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}},$$

where $SS_{res} = \sum_i (y_i - \hat{y}_i)^2$, $SS_{tot} = \sum_i (y_i - \bar{y})^2$, $\hat{y}_i$ is the model prediction, and $\bar{y}$ is the mean of y_i .

A. $R^2 = 1$

B. $R^2 = 0$

C. $R^2 < 0$

4. What are the main difference between perceptron and logistic regression?

5. What are support vectors and why are they important?

6. What is the main advantage of Physics-Informed Neural Networks compared to ordinary neural networks?